

INVESTIGATION REPORT

Issue 558: Does minstret count an mret executed in debug mode? (closed bug, replayed)

openhwgroup/cv32e40x

Document	RTL-TRIAGE-CV32E40X-558
Disposition	Bug confirmed, fix verified
Record date	2026-09-11
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40x#558
Issue type	Bug report from an integrator or user
Affects	cv32e40x
Configuration	X_EXT=0; pre-fix RTL c27b73db (2022-05-27) vs master
Disposition	Bug confirmed, fix verified
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	5 cited, 3 present, 2 not retained

Disposition meaning. The reported behaviour was reproduced, and the checks listed under Evidence pass against the corrected design.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

Confirmed on the pre-fix RTL and verified fixed on master. Property: while the core is in debug mode, an mret retiring in WB must not produce a minstret count event. On RTL c27b73db (2022-05-27, before the issue was opened) bounded model checking finds a 10-cycle counterexample: debug_req_i at cycle 4, debug mode entered at cycle 6, an mret (0x30200073) fetched from the debug halt address, decoded at cycle 7, retiring in WB at cycle 9 with the minstret event asserted. Reachability probes confirm every intermediate state (debug mode at 7, mret in ID in debug at 8, mret in WB in debug and count-in-debug at 10). On master the same property is proven for all time (abc pdr, 15 s) and holds in a 40-cycle BMC; the cover 'mret retires in WB while in debug mode' is unreachable because the decoder now flags mret in debug mode as an illegal instruction, so it never reaches WB as an mret. Comparison with the maintainers' fix (PR #567, merged 2022-06-03, read only after the proofs): decoder marks mret illegal when ctrl_fsm_i.debug_mode is set, and the controller drops the ID-stage jump to the debug exception address (PC_MRET only). This is exactly the mechanism the counterexample isolates (an mret that is allowed to decode and retire in debug mode), and the same location the issue author proposed.

3 Conditions

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Checker used

checker_558.sv, checker_558_id.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	counterexample	formal/prefix_bmc/job/engine_0/trace.vcd	not kept	pre-fix RTL: assertion fails at step 10 (23 s); rendered in cex_prefix_minstret.png
E2	proof	formal/prefix_probe/sby.log	present	pre-fix RTL: covers debug_mode (7), mret in ID in debug (8), mret in WB in debug (10), minstret count in debug (10) all reached
E3	proof	formal/master_prove/sby.log	present	master: abc pdr unbounded PASS (15 s): no minstret count for mret in debug mode
E4	proof	formal/master_bmc/sby.log	present	master: BMC depth 40 PASS
E5	diff	cores/cv32e40x_hist (git show f45ed637)	not kept	maintainers' fix PR #567: decoder ILLEGAL_INSN for mret in debug; controller PC_MRET only. Read after proving.

Paths are relative to runs/e40x-558. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.
- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- diff. Source or configuration difference.

Artefact(s) E1, E5 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

7 Limitations and scope

- The fix is verified against the checks listed under Evidence only. This is not a regression sign-off and does not cover the wider test suite.
- This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

8 Provenance and measurement

Run directory	runs/e40x-558
Title source	Title and configuration read back from reports/issues/cv32e40x_issue_558.pdf, the previously published report for this issue.
Inputs shown	title, body (which itself proposed the fix)
Human record	PR #567 (2022-06-03): decoder marks mret in debug mode illegal; controller no longer jumps to debug exception from ID
Assessed confidence	0.92. This is the investigator's own estimate. It is not a measurement and not a reliability figure.
Iterations	3

Measured tool and step time

Step	Measured
read issue + RTL (controller, decoder)	7.0 min (420 s)
checker + master cover/BMC	5.5 min (330 s)
wrong pre-fix checkout (already fixed) diagnosed	15.0 min (900 s)
true pre-fix clone + wrapper	4.0 min (240 s)
pre-fix BMC (CEX) + probes	62.0 s
master unbounded proof	15.0 s
compare with fix PR	2.0 min (120 s)
report	10.0 min (600 s)
Total	44.8 min (2687 s)

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

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